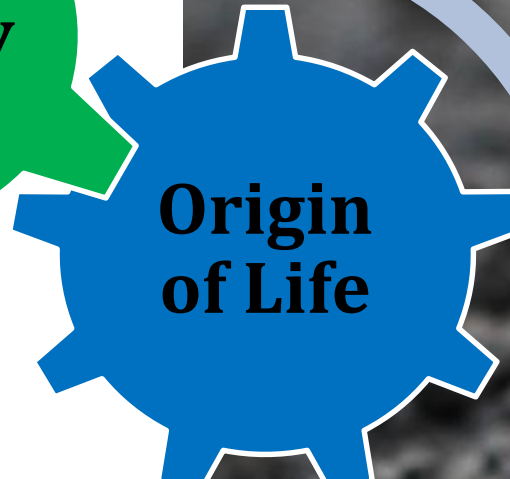
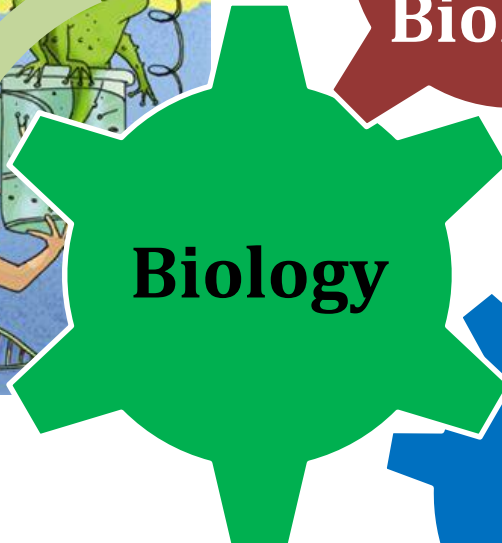
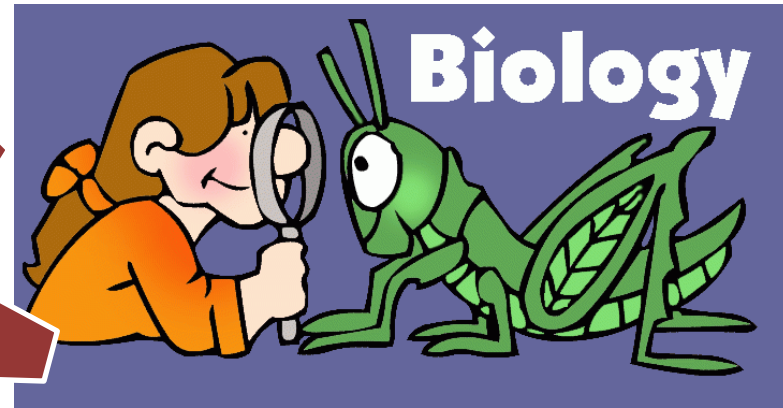
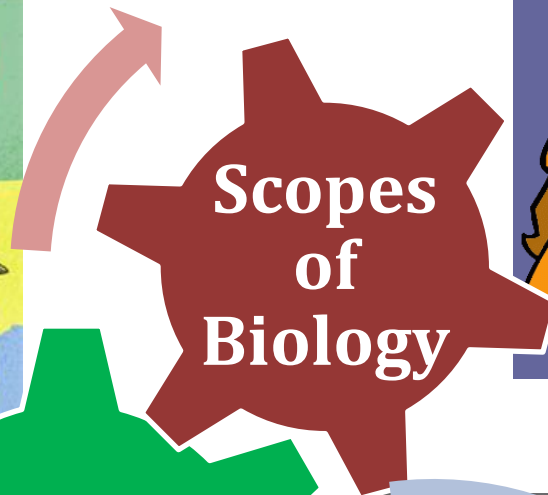
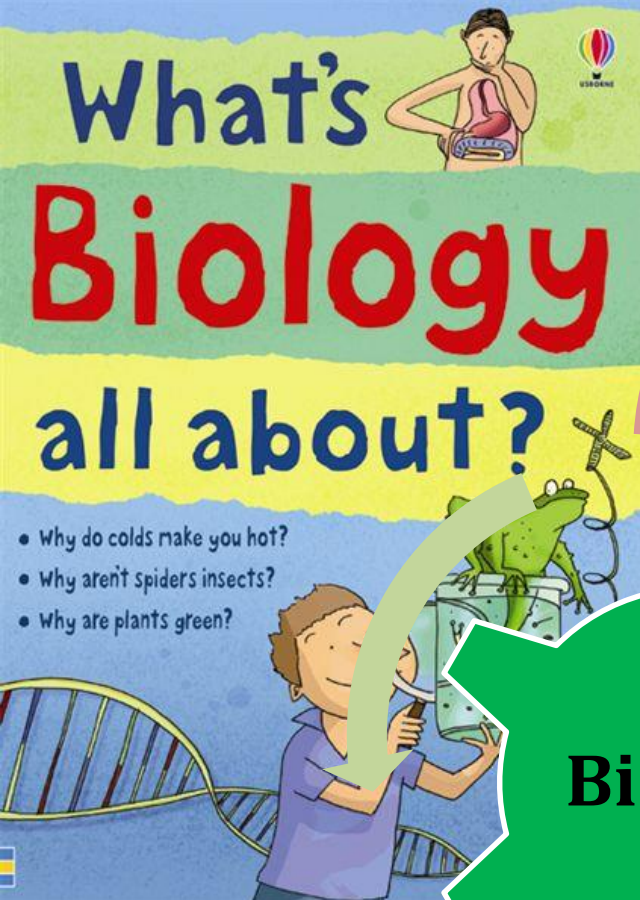


BIO 103: Introduction to Biology

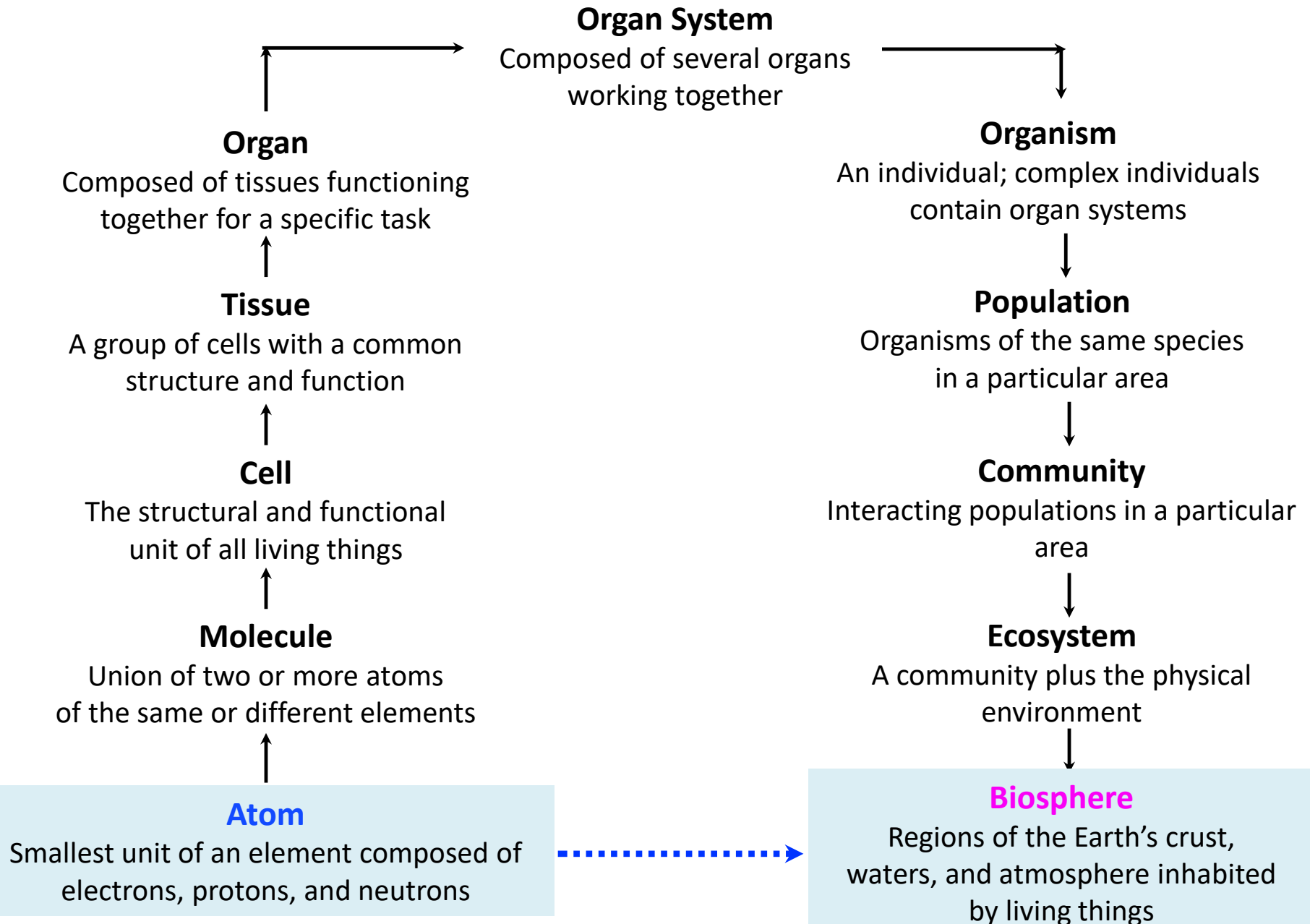
Lecture 1-2

Md. Rakibul Islam, PhD

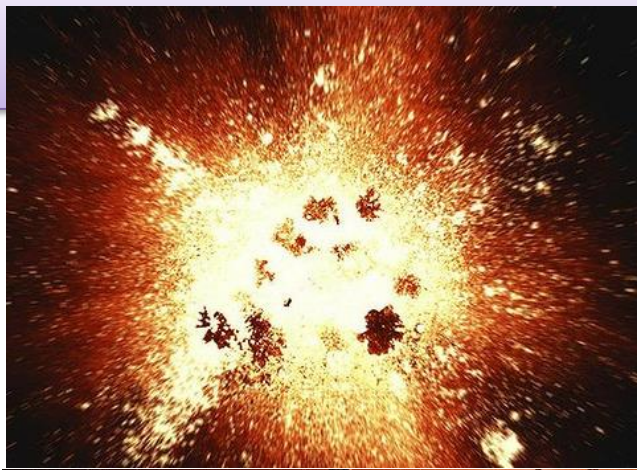


It can be defined as -
studying the science of life and living organisms.

Levels of biological organization



Origin of Life



Big Bang Theory: Origin of Earth

Explosion from a pin sized to a Huge Universe around 13.7 billion years ago. Among all Planets, Earth only contains life.

Among all Planets, Earth only contains life.



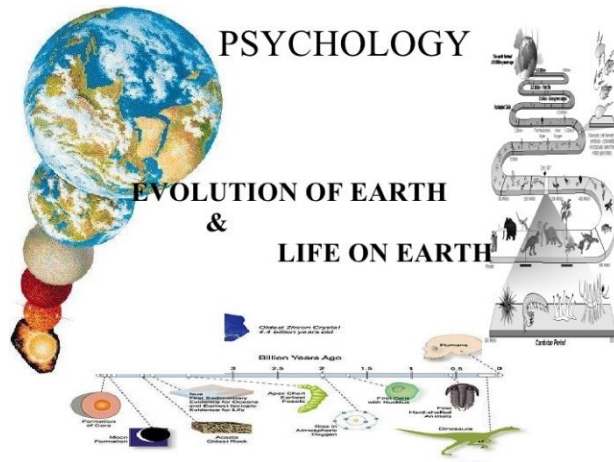
Earth without life

High temperature, poisonous gases, no free oxygen/ water. Life evolved from chemical reactions and gradual changes.

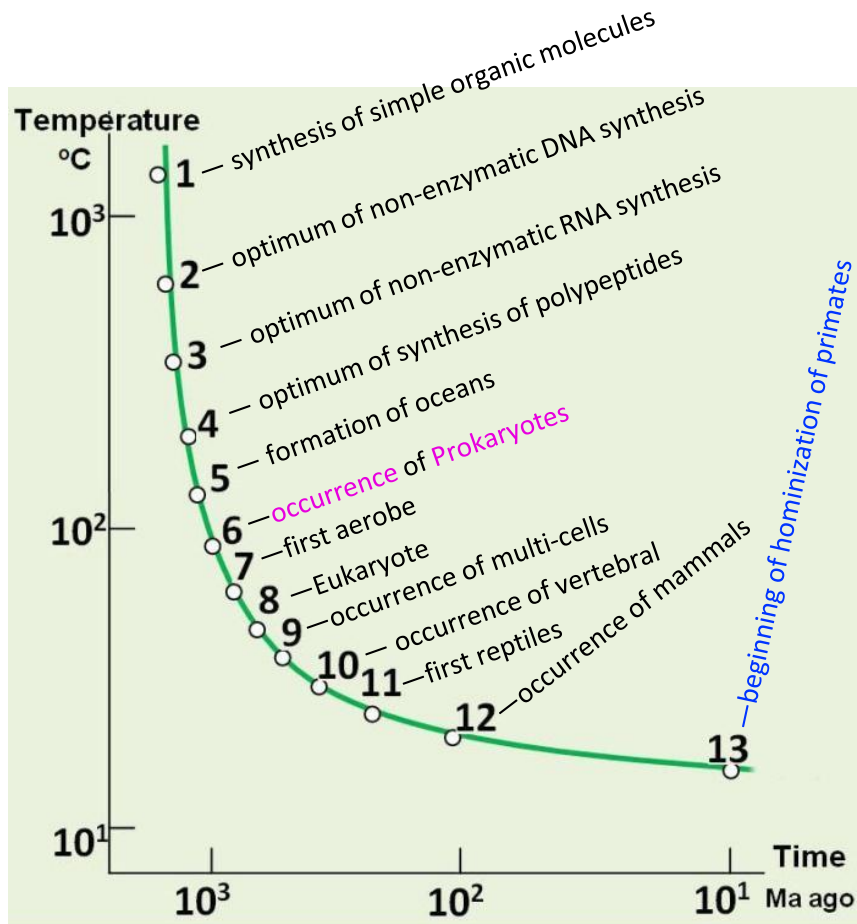


Evolution of life

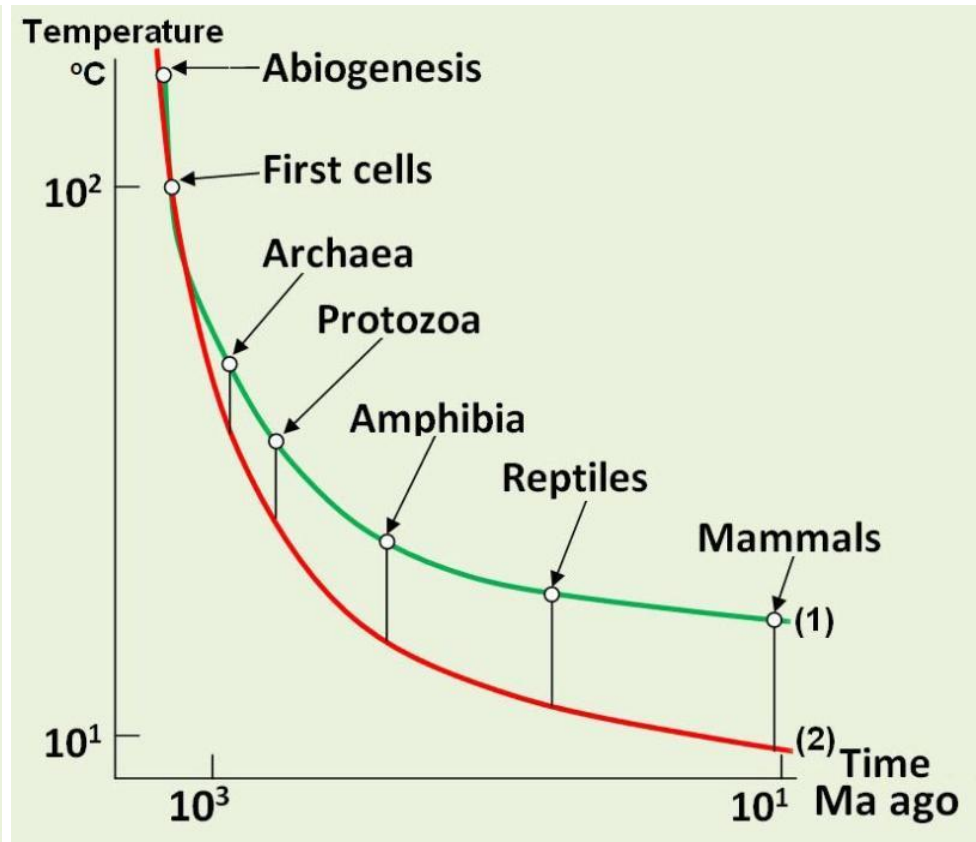
Life begins when the cell forms. Single to multi-cellular organisms. Survival of the fittest.



Life origin on the Earth



Anagenesis (species formation without branching) with Temperature evolution of the surface of the Earth.



Temperature gradient, accompanying biological evolution
 (1) — optimum of the internal medium of an organism,
 (2) — average temperature of the surface of the Earth.

Scientific Discipline

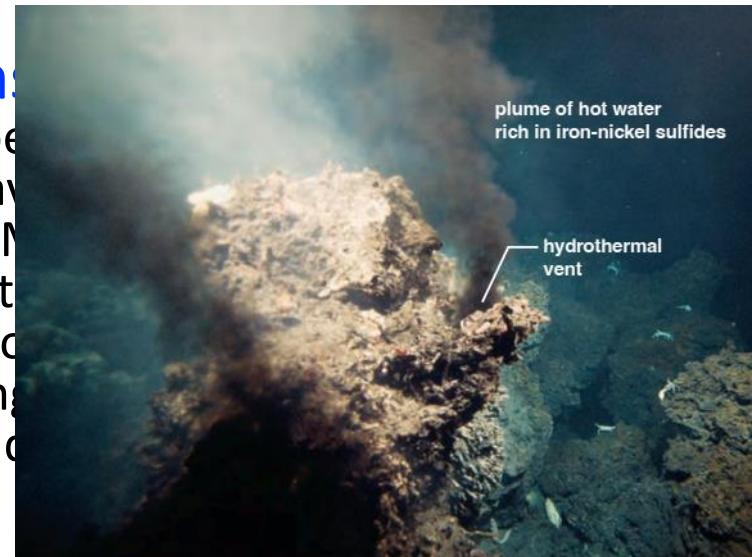
The branches of science (scientific fields or scientific disciplines) are commonly divided into **three major groups**:

- 1. Formal sciences:** the study of formal systems, such as those under the branches of *logic and mathematics*, which use an **a priori** (based on theoretical deduction), as opposed to empirical, methodology. [Logic, Data Science, Information Science, Mathematics, Statistics, Systems theory, Decision theory, Theoretical computer science]
- 2. Natural sciences:** the study of natural phenomena (including cosmological, geological, physical, chemical, and biological factors of the universe). Natural science can be divided into **two main branches**: **physical science** (Physics, Chemistry, Earth science, space science) and **life science** (Biochemistry, Microbiology, Botany, Zoology, Ecology).
- 3. Social sciences:** "science of society" is the study of human behavior in its *social and cultural aspects*. It is more likely to use qualitative assessments to drive investigations and findings.

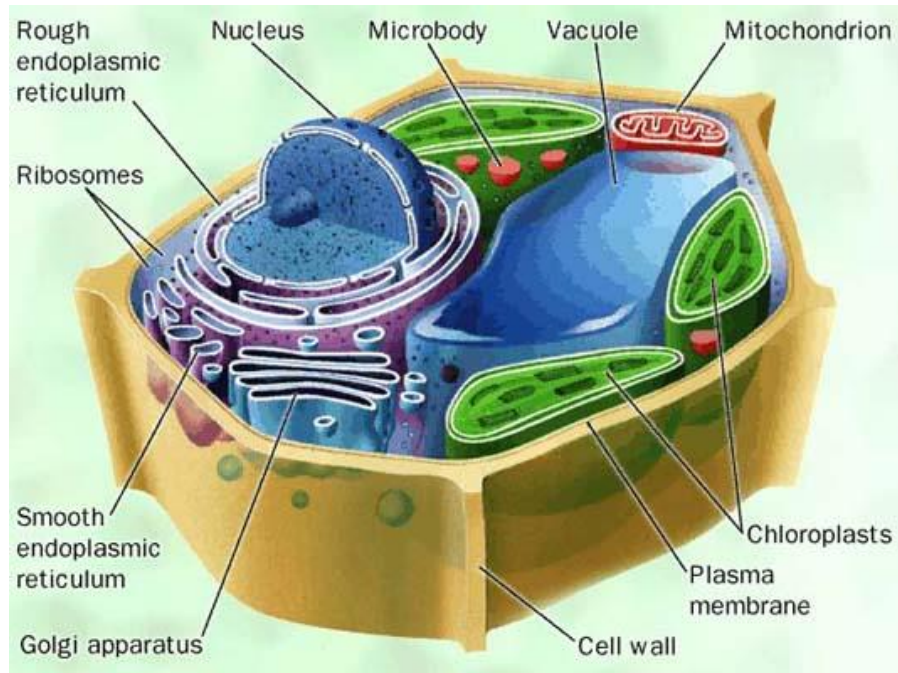
Monomers Evolve

Several hypotheses suggest how organic monomers could have evolved, and these are that:

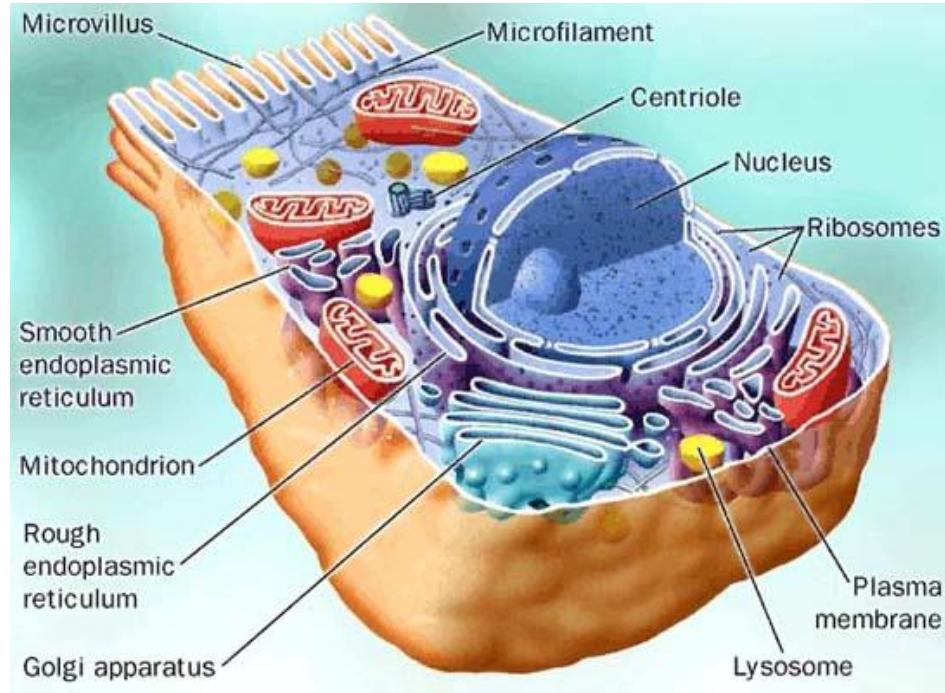
1. **Monomers came from outer space** - comets and meteorites have constantly pelted the Earth throughout history. These organic molecules could have seeded the chemical origin of life on early Earth. Others even hypothesize that bacterium-like cells evolved first on another planet and then were carried to Earth.
2. **Monomers came from reactions in the atmosphere-**
Monomers could have been produced from early atmospheric gases in the presence of strong energy sources. Stanley Miller performed a famous experiment in 1953 .
3. **Monomers came from reactions**
inert nitrogen gas (N_2) would have been in the atmosphere, ammonia (NH_3) would have been in the hydrothermal vents on the ocean floor. Water seeping through the vents is heated to $350^\circ C$, and when it spews back out, it cools and reacts with metal sulfides that can act as catalysts to change the water. Sulfides will react and amino acids will form peptides. Sulfides under ventlike conditions.



What is Biology? (Cont.)



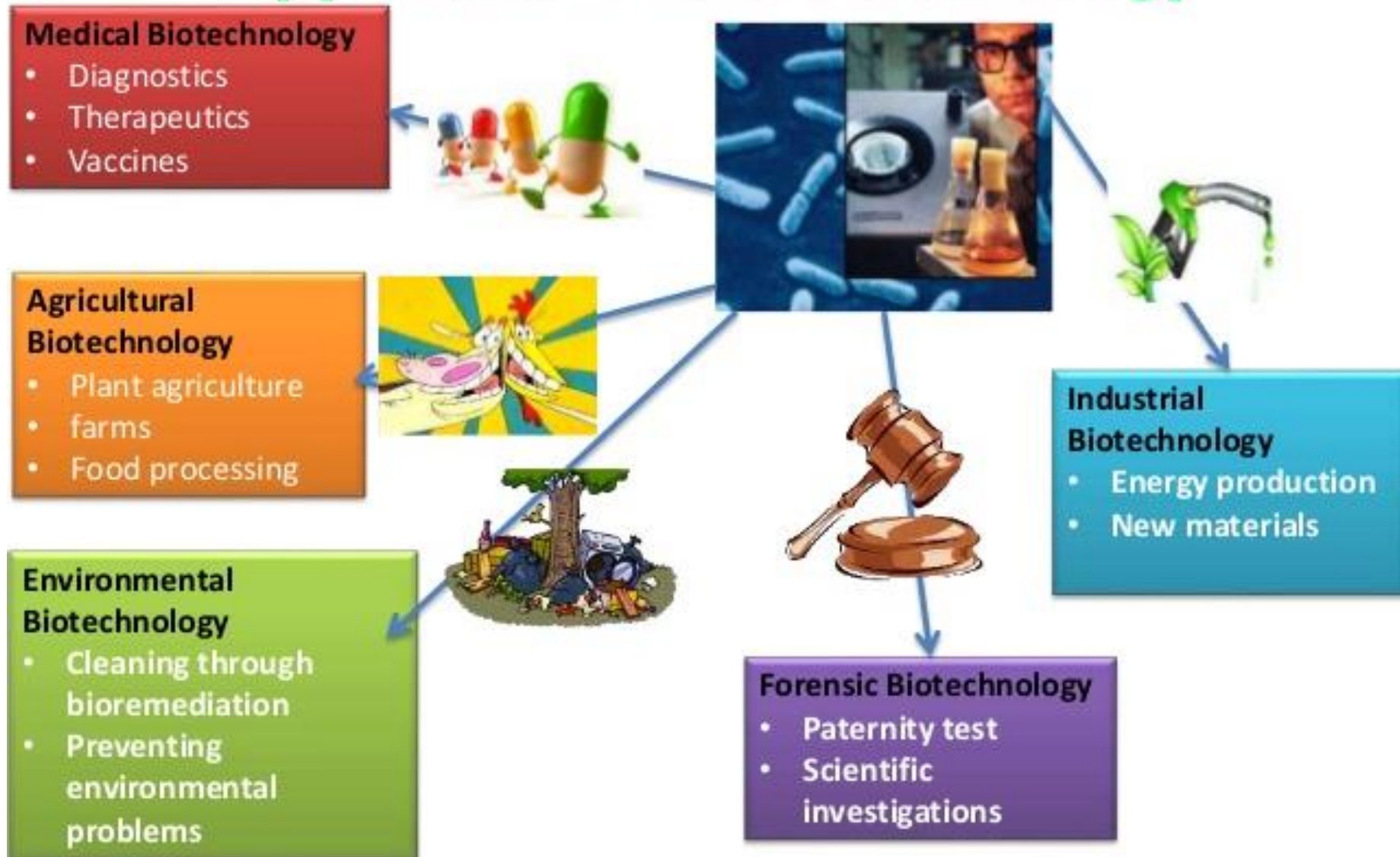
Plant cell



Animal cell

Scopes of Biology: (Cont...)

Applications of biotechnology



Characteristics of living things

There are **nine** characteristics of living things:

1. The need for energy
2. Movement
3. Cellular structure and organization
4. Growth and development
5. Maintenance and repair
6. Reproduction
7. Response to stimuli
8. Variation and adaptation
9. Metabolism

1. Need for Energy

Every living organism **needs the energy to run many activities.**

The **sun** is the universal energy source.

1. **Primary Producers:** Plants and green organisms --> produce food from sun
2. **Primary consumers:** Animals and non-green organisms--> get food from Primary producers
3. **Secondary consumers:** Get food from both Producers and Primary consumers
4. **Tertiary consumers:** Lives on primary and secondary consumers.

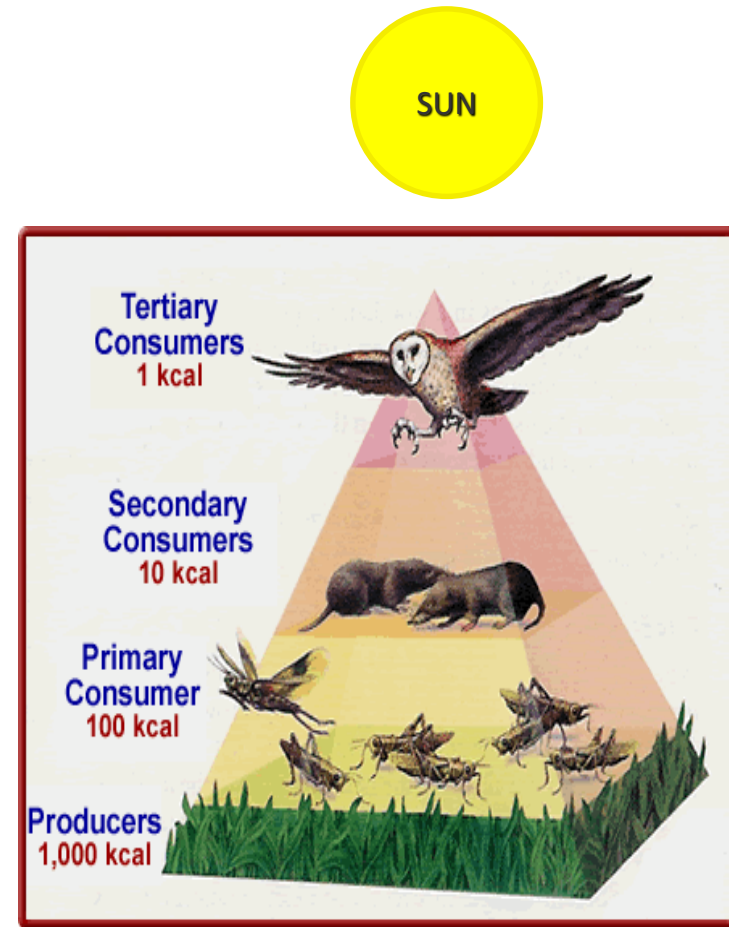


Figure: Energy Pyramid

Why.....??

- Movement
- Growth & development
- Repair and maintenance
- Response to stimuli
- Variations and Adaptation
- Reproduction
- Metabolism etc.

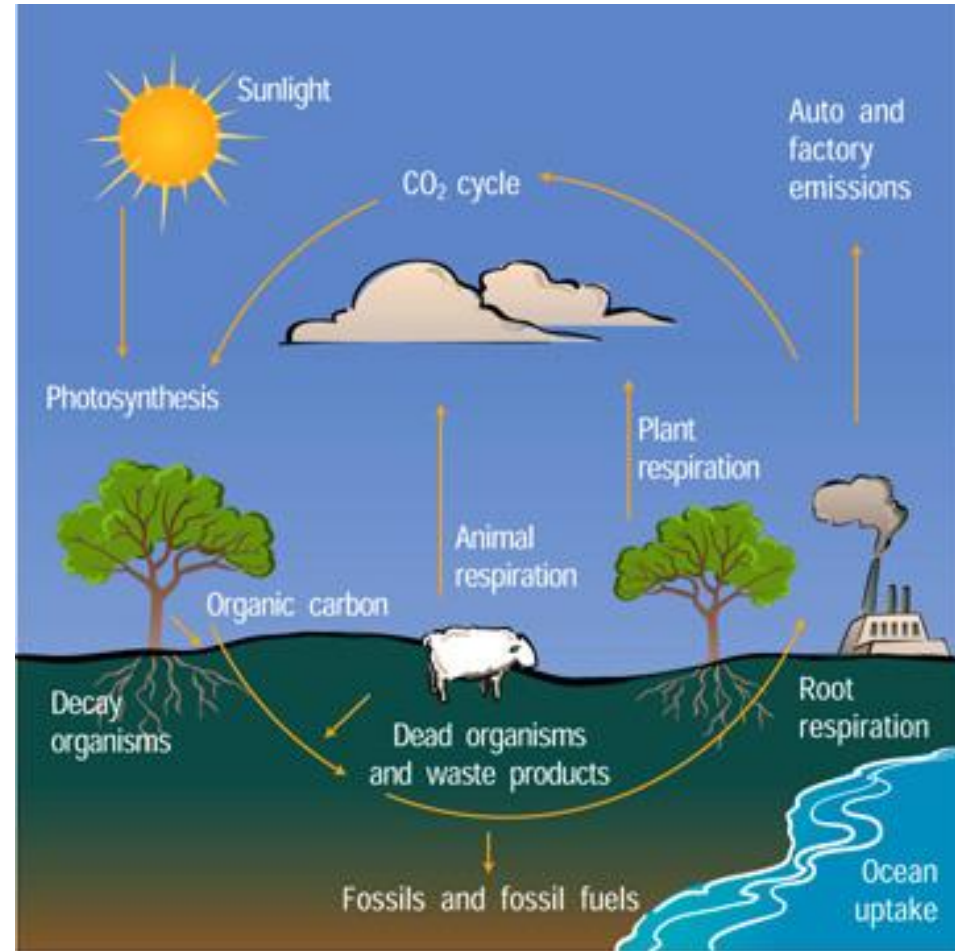


Figure: The Energy Cycle

The energy cycle describes the **interactions between energy sources within the Earth's environment.**

2. Movement (locomotion)

- i. **Animal:** Most obvious/visual
- ii. **Plant:**
 - Opening of buds
 - Turning of leaves toward sun
 - *Mimosa pudica* (the sensitive plant)
 - 'Sundew' of northern bogs: Carnivorous



Figure: 'sundew' of northern bogs.



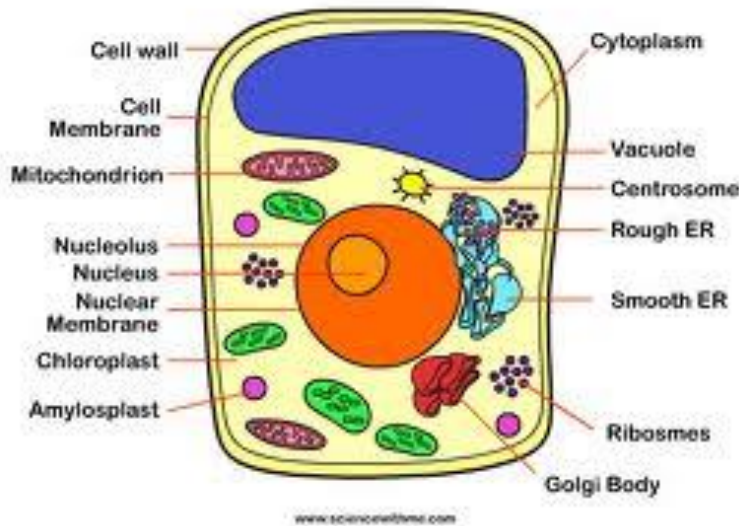
Figure: *Mimosa pudica*

Thigmotropism is when a plant will respond to touch from an external source. The plant will move toward or away from the contact stimulus.

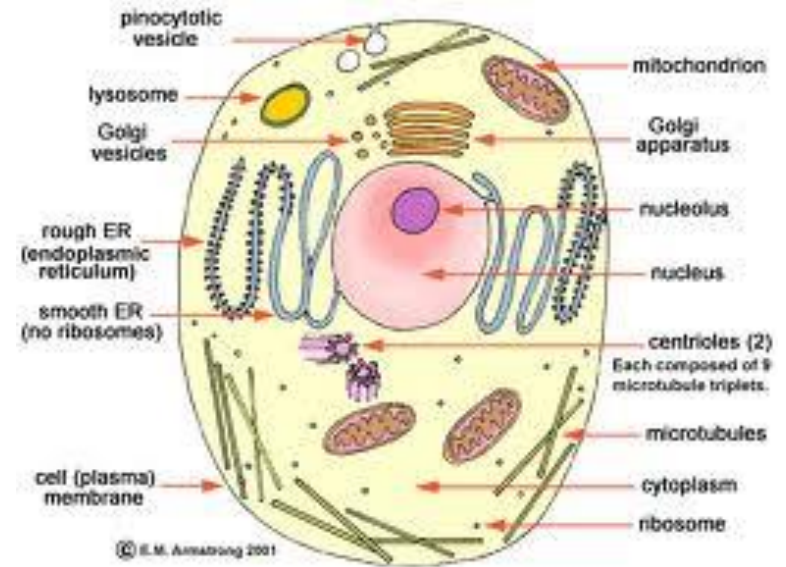
- self-propulsion by the organism.
- can be swimming, running, flying, walking etc.
- microorganisms also can propel by their own locomotion mechanism.
- But not ALL changes of location are called locomotion. e.g. wind moves the branches of trees!



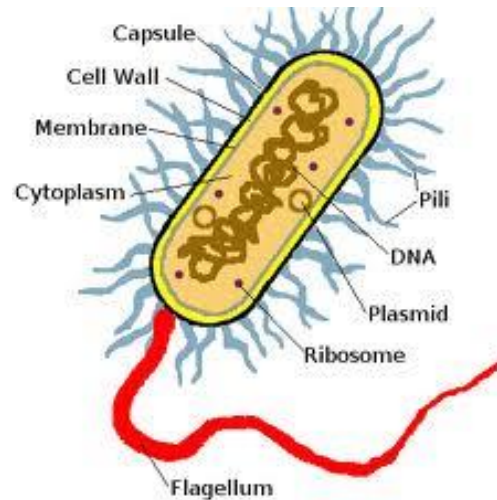
3. Cellular Structure & Organization



Plant Cell



Animal Cell



Bacterial Cell

Unicellular and multicellular organisms.

All Living things are made up of **CELLS**

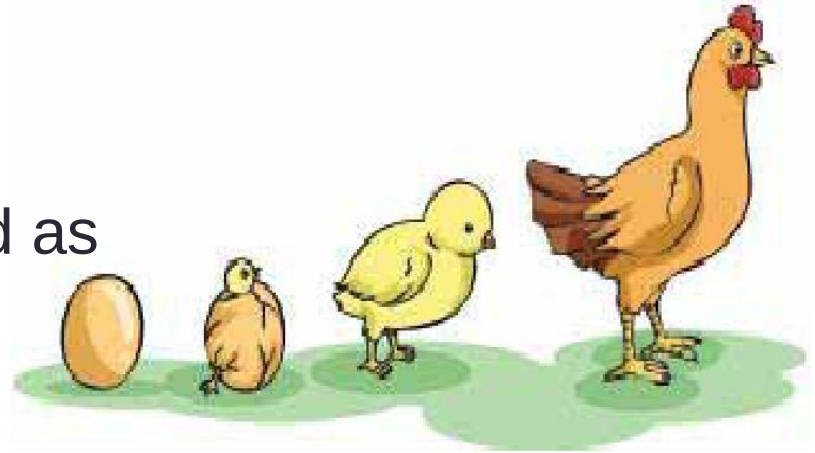
Protoplasm

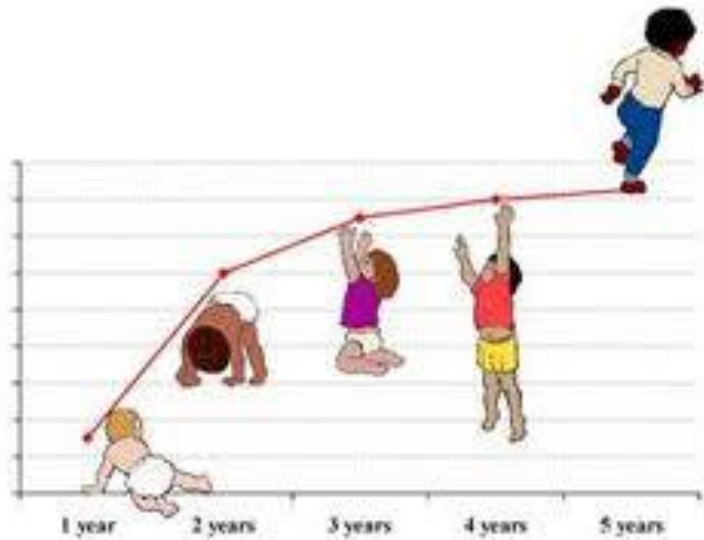
- Complex mixture of substances
- All living cells have protoplasm
- Made up of-- carbohydrates, fats, proteins, water etc.
- Ensures the living condition.
- Composition is different in different organisms.
- Even in same organism, protoplasm differs from one part to another --> always changing.
- Living things can organize materials into protoplasm and organize protoplasm to make cells and can organize cells to make a total living thing. **HENCE**, for this organizing capacity, they are called “**Organisms**”.

4. Growth & Development

Growth:

- Getting larger is not only termed as growth.
- Collection of things are not the growth.
- *By organizing materials, living things make special kinds of protoplasm, reproduce cells/increase the cell number, and go from one state to another.*





Development:

- A series of changes that take place as an organism grows toward final form, is called **development**.
- By development, an organism **becomes a unique living thing** with specialized parts which are different from another.

5. Maintenance & Repair

- ❑ Part of growth and development
- ❑ Appears to have stopped growth, but actually system runs as 'maintenance and repair'
- ❑ Examples:
 - Replacing dead skin cells.
 - Healing of cut fingers/skin.
 - Tail recovery of house lizard.
 - Replenish Red blood cells in 120 days. ..etc



6. Reproduction

Only living things can produce offspring similar to themselves.

Types of Reproduction: There are two types of reproduction:

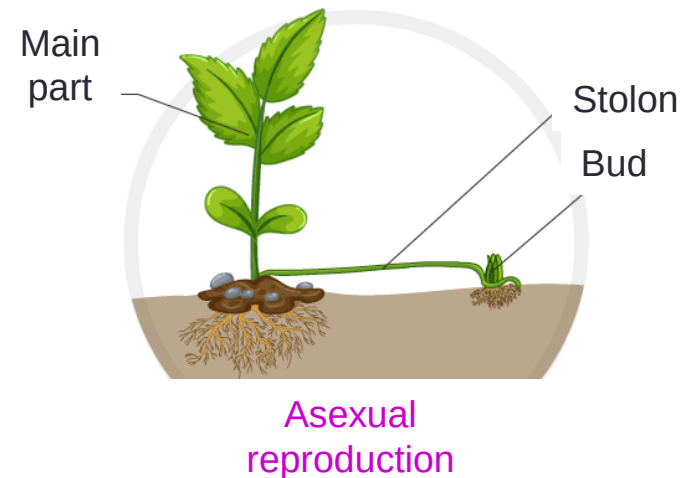
- **Asexual Reproduction** - does not involve the fusion of gametes i.e. Binary Fission, Budding, Sporogenesis,
- **Sexual Reproduction** - involves the production of offspring by the fusion of male and female gametes.

❑ The process can be different—

- Laying eggs
- Giving birth
- Plant seeds develop into plants etc.

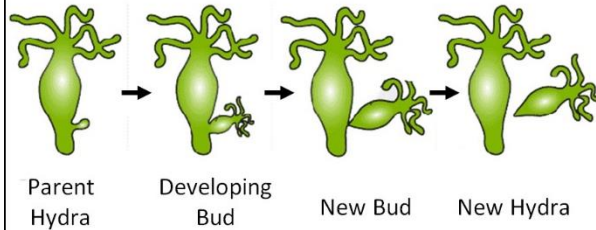
❑ The law of nature is:

❑ **“life produces life” and “like produces like”**

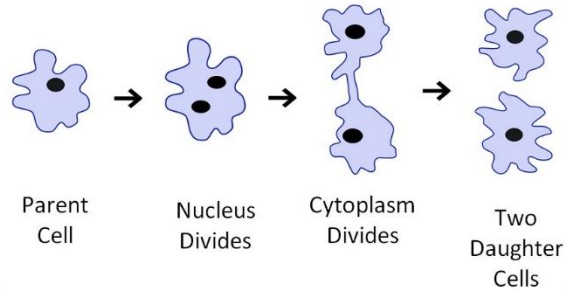


Asexual Reproduction Takes place by different methods

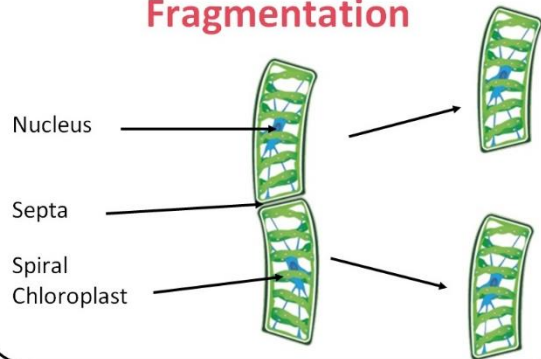
Fission



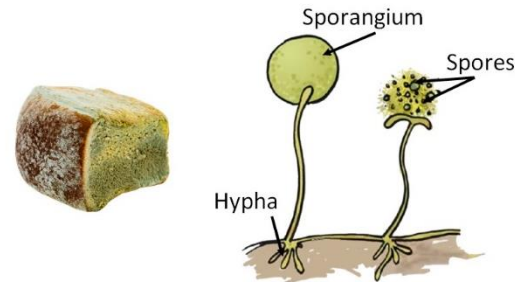
Budding



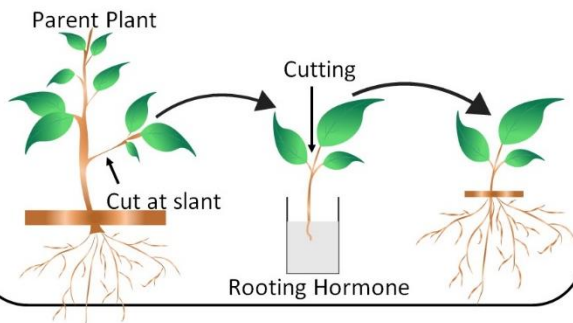
Fragmentation



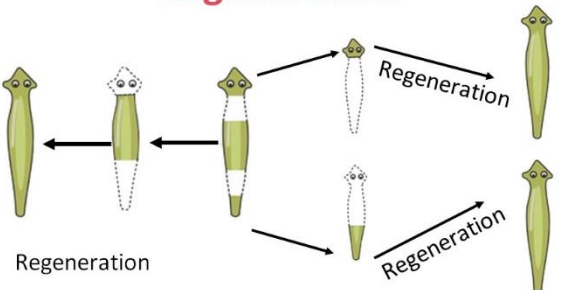
Spore Formation



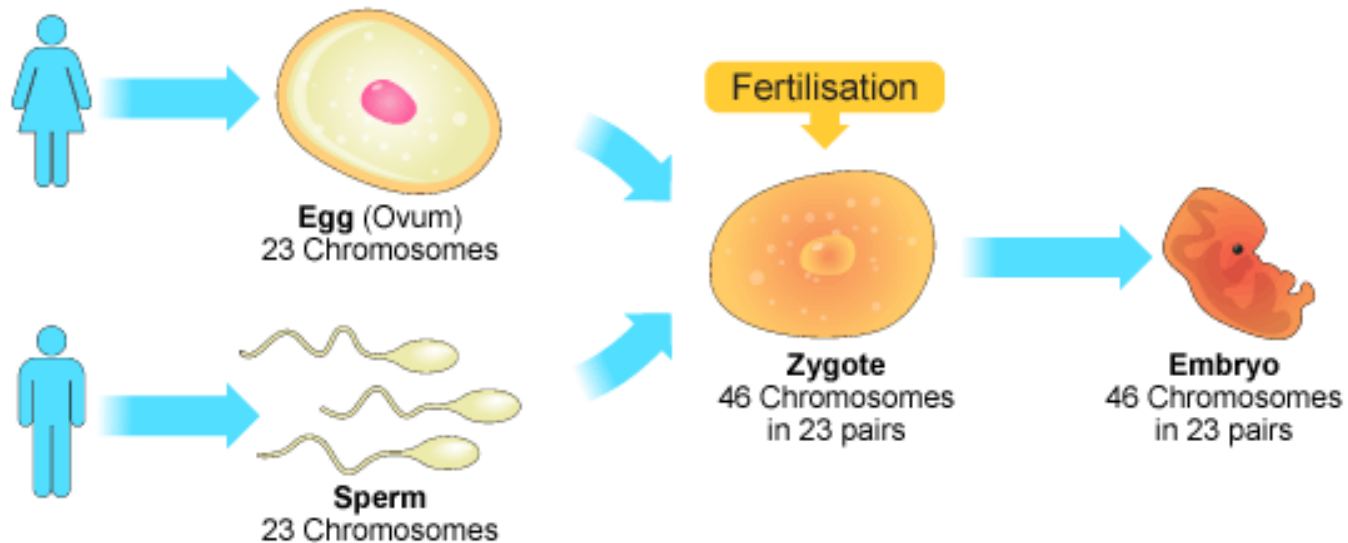
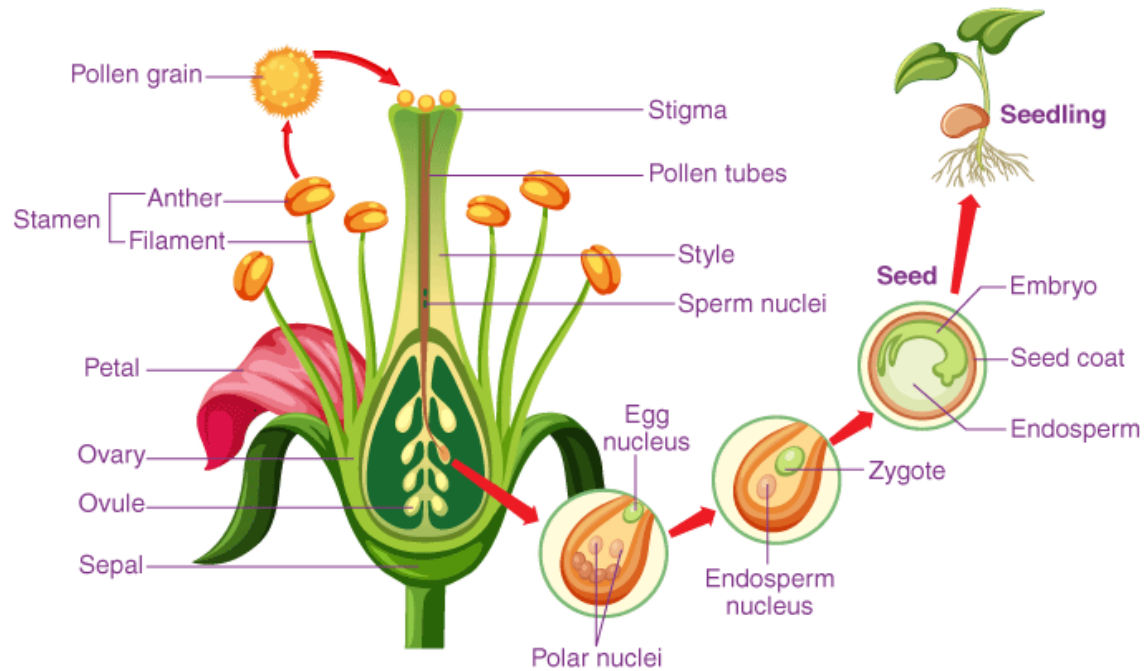
Vegetative Propagation



Regeneration



Sexual Reproduction



Differences between Sexual and Asexual reproduction are:

Sexual	Asexual
Two parents take part	Single parent
Variation occurs in offspring	Offspring are genetically identical to each other and to their parent
Fertilization takes place	No internal fertilization or external fertilization takes place
Gametes are involved	No gamete is involved
Mixing of hereditary material takes place	No mixing of hereditary material takes place

7. Response to Stimuli

Animal:

- Have nervous system
- Use eye, nose, and ear to respond

Plants:

- Do not have a nervous system
- Slow response to light, oxygen, nutrients etc

Microorganisms:

- Response to nutrients, environments
- Even single-celled organisms can respond

8. Variation and Adaptation

Variation:

- Change occurs as a result of characteristic.
- Most variations do not affect an organism's chances of survival
- Give an organism a better chance of surviving in a changing environment



Adaptation:

The process by which a certain type of organisms becomes better suited to survive in its environment is called **adaptation**.

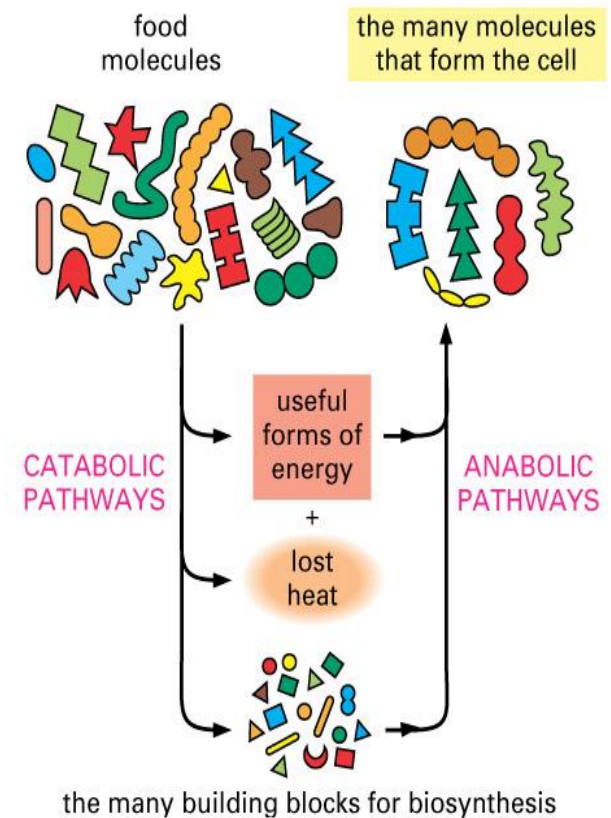
e.g.: Hibernation, Mimicry (The viceroy butterfly mimics the monarch butterfly by coloration. The monarch butterfly has a bitter taste to predators and predators avoid them.

9. Metabolism

- The exchange of matter and energy between the organisms and within an organism.
- Metabolism is the **sum of all the processes occurring** in an organism.
 - Ingestion
 - Digestion
 - Assimilation
 - Respiration
 - Excretion

Metabolism has two distinct phases --

1. **Anabolism:** Building up
2. **Catabolism:** Breaking down



Binomial nomenclature

- After classifying, Linnaeus decided to name all the organism.
- Every organism's name is consist of **two words** → Binomial Nomenclature
- He decided to write all names in *Latin*.

Example:

• **Common name:** Human

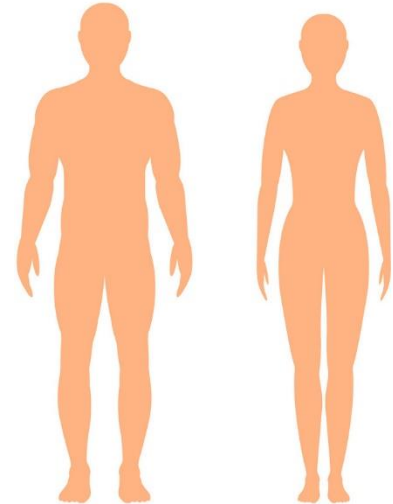
• **Scientific name:**

Homo sapiens (Italic if printed)

Homo sapiens (separate underlines if handwritten)

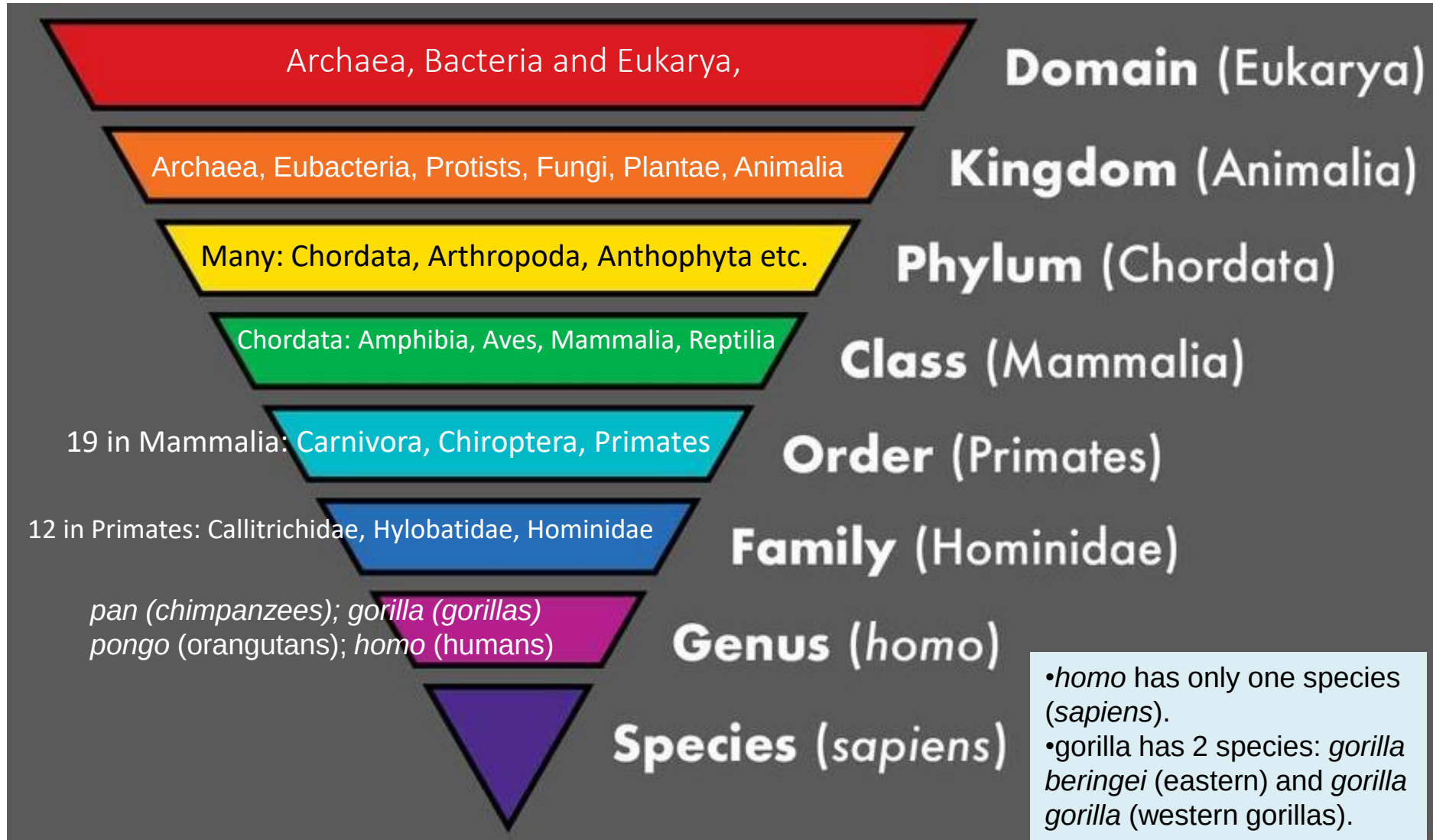
Common name: Cat

Scientific name: *Felis domesticus*



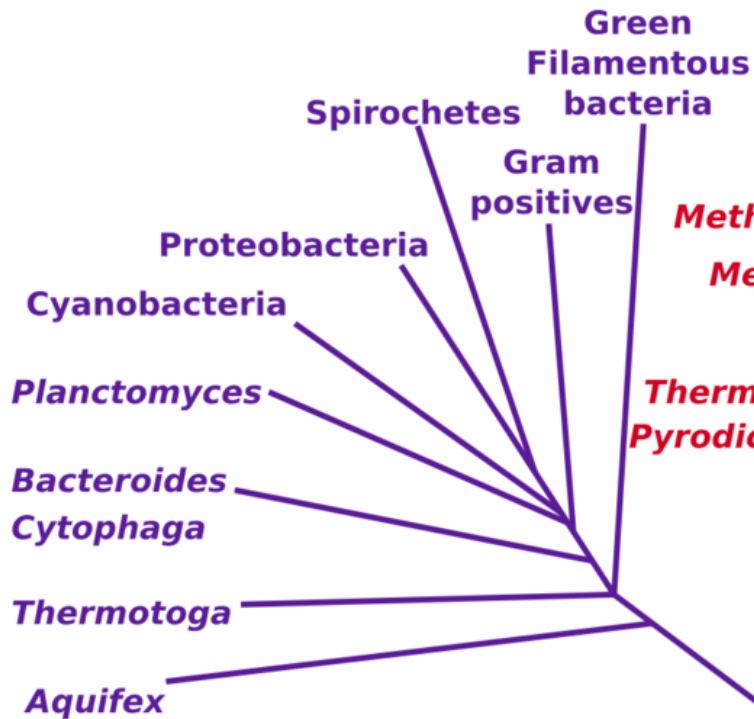
Taxonomy of living things

There are 7-8 main ***taxa*** or classification groups.

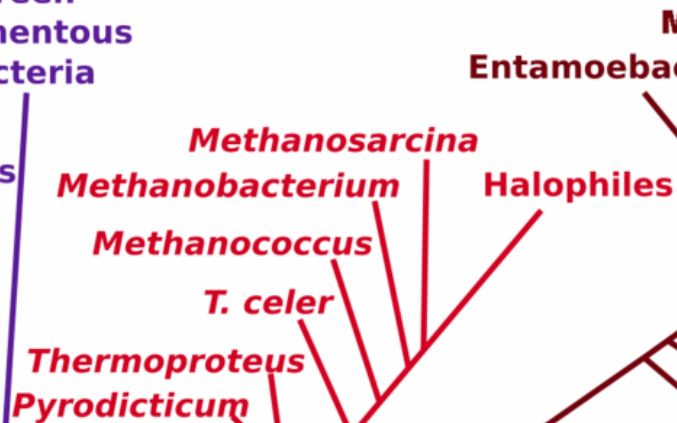


Phylogenetic Tree of Life

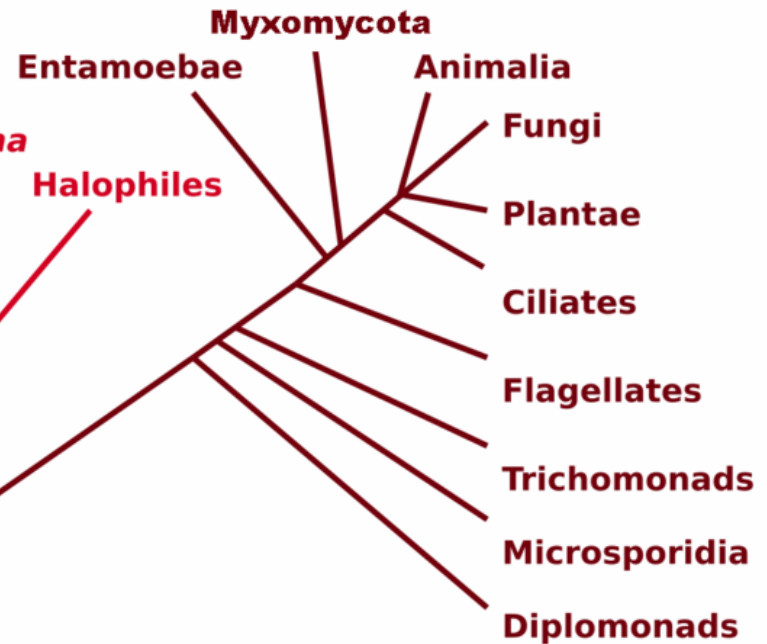
Bacteria



Archaea



Eucarya



Main classification groups (taxa)

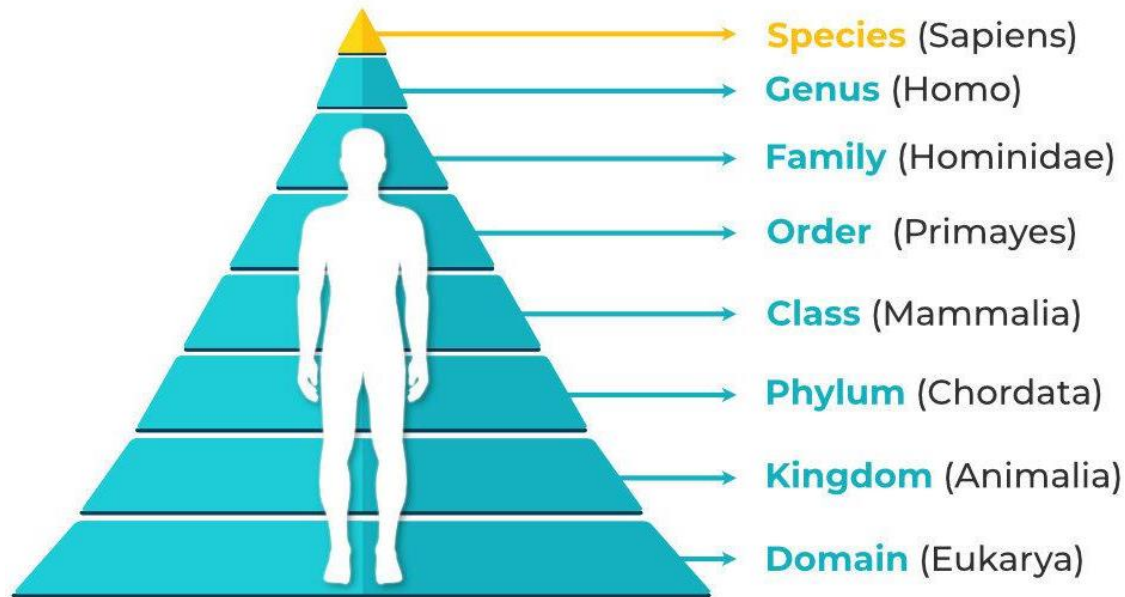
Example:



Taxon	Human	Gorilla
Kingdom	Animalia	Animalia
Phylum or Division	Chordata	Chordata
Class	Mammalia	Mammalia
Order	Primates	Primates
Family	Homonidae	Pongidae
Genus	<i>Homo</i>	<i>Gorilla</i>
Species	<i>sapiens</i>	<i>gorilla</i>

Meet your human taxonomy

Taxonomy of Human



H. sapiens

(the largest brain case of the genus *Homo*, giving us the capacity for complex speech; “*sapiens*” loosely translates as “knowing”)

We are the only living organisms in our species, with a unique set of combined characteristics from our family (bipedal), order (opposable thumbs), and genus (large brain case).



“You’re only here for a short visit. Don’t hurry, don’t worry. And be sure to smell the flowers along the way.”

—*Walter Hagen*

